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SIS №2
Development of sensor connection and actuator control circuits for an automated washing powder production line

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Almaty, 2025

Content

1. Introduction 3
2. Tables listing the equipment 4
3 Equipment selection justification 6
4. References 7

Introduction

This report presents the detailed design of the instrumentation and control systems for a fully automated washing powder production line. Building upon the foundational power distribution framework established in the previous stage, this document focuses on the low-voltage control circuits that form the nervous system of the entire manufacturing process. The primary objective is to develop a comprehensive set of schematic diagrams and justifications for the selection and integration of all field sensors and actuators with the central control unit.

The core of the automation system is the Siemens SIMATIC S7-1500 Programmable Logic Controller (PLC), which orchestrates every stage from raw material intake to final packaging. This report provides a detailed blueprint for connecting a diverse array of sensors responsible for gathering critical real-time process data. This includes Siemens SITRANS P sensors for precise level and pressure monitoring , SITRANS TS500 temperature sensors for thermal process control , and SITRANS F M flow meters for accurate liquid dosing. Furthermore, specialized instruments such as the SITRANS WS300 speed sensor for the granulator and vibration sensors for the filtration system are integrated to ensure optimal equipment performance and product quality.

In parallel, this document outlines the actuator control schematics. It details how the PLC's digital outputs manage the executive components of the production line—primarily the powerful induction motors—through a series of EKF contactors. By presenting the wiring diagrams and operational principles for both the sensory input and control output systems, this work provides a complete and practical guide for the implementation of a modern, efficient, and reliable automated production facility.

Tables listing the equipment

Table listing the equipment used for **Mixer:**

№	Device name	Type	Characteristic	Source
1	Level sensor	AI	Siemens SITRANS P DS III (7MF4033); 4-20mA. 1 to 100 bar.	link to the manual
2	Temperature sensor	AI	Siemens SITRANS TS500 (with transmitter TH420); 4-20mA. –50 ... +400 °C.	link to the manual SITRANS TS500 link to the manual SITRANS TH420
3	Flow sensor	AI	Siemens SITRANS F M MAG 5100 W; 4-20mA. 0 to 10 m/s. 0.2 % ±2.5 mm/s. Max. 16 bar. From -40 to 70 °C.	link to the manual
4	Temperature relay	DI	EKF RT-820M (with RT801); 24...240 V DC/AC, –25 ... +130 °C.	link to the manual
5	Contactor	DO	EKF KME 50A ; 24V DC 1NO 1NC PROxima, 3-pole.	link to the manual

Table listing the equipment used for **Disperser:**

№	Device name	Type	Characteristic	Source
1	Level sensor	AI	Siemens SITRANS P DS III (7MF4033); 4-20mA. 1 to 100 bar.	link to the manual
2	Pressure sensor	AI	Siemens SITRANS P DS III (7MF4033); 4-20mA. 1 to 100 bar.	link to the manual
3	Temperature sensor	AI	Siemens SITRANS TS500 (with transmitter TH420); 4-20mA. –50 ... +400 °C.	link to the manual SITRANS TS500 link to the manual SITRANS TH420
4	Temperature relay	DI	EKF RT-820M (with RT801); 24...240 V DC/AC, –25 ... +130 °C.	link to the manual
5	Contactor	DO	EKF KME 40A ; 24V DC 1NO 1NC PROxima, 3-pole.	link to the manual

Table listing the equipment used for **Granulator:**

№	Device name	Type	Characteristic	Source
1	Speed sensor	HSC	Siemens SITRANS WS300; 24V pulses. 0.5 to 2000 rpm.	link to the manual
2	Flow sensor	AI	Siemens SITRANS F M MAG 5100 W; 4-20mA. 0 to 10 m/s. 0.2 % ±2.5 mm/s. Max. 16 bar. From -40 to 70 °C.	link to the manual
3	Temperature relay	DI	EKF RT-820M; 24...240 V DC/AC, –25 ... +130 °C.	link to the manual
4	Contactor	DO	EKF KME 65A ; 24V DC 1NO 1NC PROxima, 3-pole.	link to the manual

Table listing the equipment used for **Dryer:**

№	Device name	Type	Characteristic	Source
1	Temperature sensor (input)	AI	Siemens SITRANS TS500 (with transmitter TH420); 4-20mA. –50 ... +400 °C.	link to the manual SITRANS TS500 link to the manual SITRANS TH420
2	Temperature sensor (output)	AI	Siemens SITRANS TS500 (with transmitter TH420); 4-20mA. –50 ... +400 °C.	link to the manual SITRANS TS500 link to the manual SITRANS TH420
3	Humidity sensor	AI	Siemens QFM2171; 4-20mA. IP54.	link to the manual
4	Temperature relay	DI	EKF RT-820M (with RT801); 24...240 V DC/AC, –25 ... +130 °C.	link to the manual
5	Contactor	DO	EKF KME 40A ; 24V DC 1NO 1NC PROxima, 3-pole.	link to the manual

Table listing the equipment used for **Filtration System:**

№	Device name	Type	Characteristic	Source
1	Vibration sensor	AI	IFM VTV122; 4-20mA.	link to the manual
2	Temperature relay	DI	EKF RT-820M (with RT801); 24...240 V DC/AC, –25 ... +130 °C.	link to the manual
3	Contactor	DO	EKF KME 40A ; 24V DC 1NO 1NC PROxima, 3-pole.	link to the manual

Table listing the equipment used for **Packaging Machine:**

1	Temperature relay	DI	EKF RT-820M (with RT801); 24...240 V DC/AC, –25 ... +130 °C.	link to the manual
2	Contactor	DO	EKF KME 40A ; 24V DC 1NO 1NC PROxima, 3-pole.	link to the manual

Equipment selection justification

Temperature Relay (EKF RT-820M): A simple relay that opens the circuit when a set temperature is reached. It is used as protection against motor overheating.

1. Mixer

1) Level Sensor (Siemens SITRANS P DS III (7MF4033)): Measures the pressure of the material column and converts it into a level signal (4-20 mA). It is needed for precise dosing of components according to the recipe.

2) Temperature Sensor (Siemens SITRANS TS500 (with TH420 transmitter)): The resistance temperature changes with temperature, and the TH420 transmitter generates a 4-20 mA signal. It monitors the temperature of the mixture to prevent overheating and damage to the components.

3) Flow Sensor (Siemens SITRANS F M MAG 5100 W): An electromagnetic sensor measures the flow rate of the liquid. Ensures precise addition of liquid ingredients.

4) Contactor (EKF KME 50A; 24V DC): An electromagnetic switch that switches the 380V motor on/off with a 24V signal from the PLC. Safely controls the powerful mixer motor.

2. Disperser

1) Level and Pressure Sensors (Siemens SITRANS P DS III (7MF4033)): Measure pressure to determine the system level and pressure. The level sensor protects the pump from dry running, and the pressure sensor protects the system from overpressure.

2) Temperature Sensor (Siemens SITRANS TS500 (with TH420 transmitter)): Measures the temperature of the mixture. Maintains the optimal temperature for dissolving additives.

3) Contactor (EKF KME 40A; 24V DC): Switches the disperser motor on/off with a PLC command. Controls the unit's drive.

3. Granulator

1) Speed sensor (Siemens SITRANS WS300): Contactlessly measures rotation speed and transmits data to the PLC via the HSC inputs, which counts 24V pulses. Controls drum rotation speed, which directly affects granule size and quality.

2) Flow sensor (Siemens SITRANS F M MAG 5100 W): Measures liquid flow. Precisely supplies the binder for granule formation.

3) Contactor (EKF KME 65A; 24V DC): Controls the most powerful motor in the line. Starts/stops the granulator drive.

4. Dryer

1) Temperature sensors (input/output, Siemens SITRANS TS500 (with TH420 transmitter)): Two sensors measure the air temperature before and after the drying chamber. Allows the PLC to evaluate drying efficiency based on temperature differences and save energy.

2) Humidity Sensor (Siemens QFM2171): Measures relative air humidity. A key parameter for determining the end of product drying.

3) Contactor (EKF KME 40A; 24V DC): Controls the fan motor and heaters. Controls the dryer actuators.

5. Filtration System

1) Vibration Sensor (IFM VTV122): Measures the vibration speed of the vibrating screen mechanism. Monitors vibrating screen operation; deviations from the norm indicate clogging or failure.

2) Contactor (EKF KME 40A; 24V DC): Turns the vibration motor on/off. Controls the vibrating screen drive.

6. Packaging Machine

1) Contactor (EKF KME 40A; 24V DC): Controls the packaging machine drives. Starts/stops the packaging.

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